

Embodied Motion as Interface: Exploring Abstract Dynamics in Hybrid Environments

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Figure 1: The visual design of FORM in Unreal Engine 5. Image credit: Wenhao Li, 2025

Abstract

As human physicality and digital realms continue to merge, our bodies can now interact directly with digital elements. This research explores how movement can be digitized and reimagined through FORM—a motion-capture interactive prototype using a Rokoko suit and Unreal Engine 5.

Users interact with 3D objects and sound via subtle gestures, generating real-time audiovisual feedback. Embracing abstraction and unpredictability, FORM positions the body as an expressive interface, fostering a dynamic dialogue between user and system. The work contributes to evolving discussions in Human-Computer Interaction (HCI).

CCS Concepts

• **Human-centered computing** → **Gestural input; Auditory feedback**; • **Computer systems organization** → **Real-time languages**.

*All authors contributed equally to the conceptual development and execution of the project. The extended abstract was written by Josephine Putri.

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Keywords

Body Movement, Embodiment, Motion Capture, Gesture Interaction, Interactive Systems, Human-Computer Interaction

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1 Introduction

The human body is increasingly used as a dynamic interface for digital interaction, extending its presence through motion capture and virtual embodiment. While systems like AΦE's Lilith.Aeon Genesis [Nakamura and Lecoq 2022] and McGregor's AISOMA [McGregor 2018] enhance choreographic potential, they often prioritize precision over spontaneity and emotion [Winship 2024; Yang 2024].

This research introduces FORM [Putri 2025], a prototype that reimagines the body as a co-creator of digital environments. Inspired by the works like Whitley's Digital Body [Whitley 2020], FORM uses real-time feedback to turn movement into generative audiovisual output. FORM explores subtle gestures and improvisational input, treating the body's subtle movements and "noise" as potential creative material.

2 Method

The project was developed through iterative play-testing using Rokoko suits and Rokoko Studio Live to stream real-time motion

data into Unreal Engine 5. Integrated with MetaSound-generated audio and visual effects, user movements shaped the form and atmosphere of the virtual environment, enabling abstract, embodied interaction.

2.1 Prototype Development Phases

The prototype progressed through four iterative phases, each enhancing responsiveness and immersion:

- **Gesture-Driven Control:** Linked bodily presence to environmental changes across varied movement styles.
- **Audiovisual Feedback:** Enabled gestures to trigger real-time changes in particles, lighting, and sound.
- **Adaptive Environments:** Explored dynamic responses of the virtual space to user movement.
- **Real-Time Integration:** Combined hardware and software for fluid interaction between user and environment.

This process also generated a growing database of full-body interaction mappings (see Table 1), guiding users in engaging with FORM's audiovisual responses through both subtle and expansive gestures.



Figure 2: A performer testing the prototype during early development. Image credit: Josephine Putri, 2025

During testing, participants engaged the audiovisual system through simple, mapped movements that generated meaningful feedback loops and encouraged exploration. Notably, a professional performer's more expressive gestures triggered unexpected responses—for instance, pulling the arms back produced a unique sound effect not seen with other users (Figure 2). This demonstrated the system's potential to accommodate diverse movement styles and interpretations.

3 Conclusion and Future Work

The project explores a body-driven alternative to traditional motion capture by emphasizing abstract interaction and digital unpredictability. Iterative testing revealed both technical challenges and the creative flexibility of real-time engines and motion capture within an artistic context. While the current system relies on mocap suits, future versions aim to enhance sensitivity to subtle gestures

Table 1: Full Body Interaction Mapping

Body Part	Movement	Effect
Full Body	Body movement	Particles follow hand direction.
	Vary speed/intensity	Flame expands or shrinks.
	Open/close arms	Light brightens or dims.
Hands	Close left hand	Gravity pulls inward.
	Thumb + index	Adjust volume.
	Thumb + middle	Control time/speed.
Legs	Open/close palms	Change pitch.
	Arms up/down	Adjust frequency.
	Widen stance	Change floor visuals.

and interpret small-scale movement more expressively. Collaborations with dancers and choreographers are planned to deepen its performative potential.

These results support the growing accessibility of motion capture. Tools like auto-rigging and Kinect allow artists to explore motion-based interaction beyond industry settings. Though precision remains a challenge, emerging marker-less systems show promise [Yang 2024]. Ultimately, creative technologies should amplify, not constrain, human expression. As AI and related tools advance, human creativity remains central—unlocking new artistic possibilities through collaboration with technology.

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